

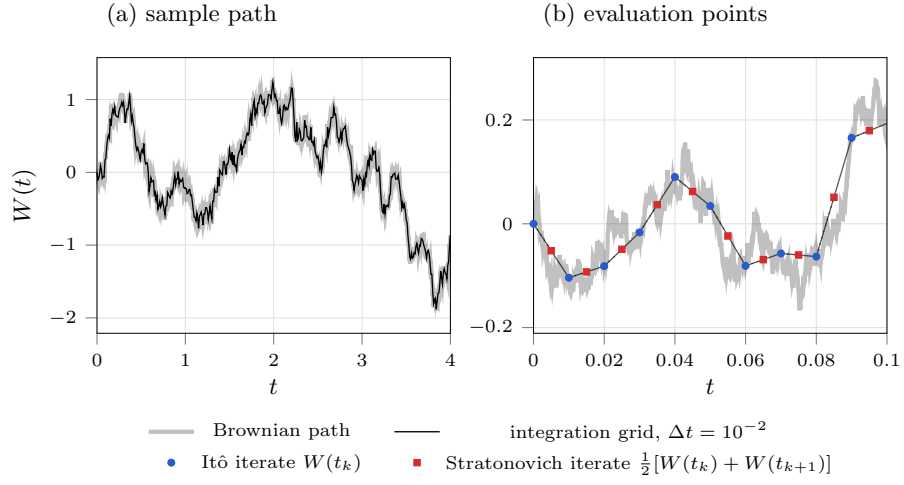


Figure 1: A Brownian path at $\Delta t = 10^{-4}$ (pale) with the integration grid at $\Delta t = 10^{-2}$ laid over it. (b) magnifies the first ten steps: the Itô rule reads W at the left end of each step, the Stratonovich rule takes the mean of the two ends, which is the midpoint of the chord and not a value of the path.

Itô versus Stratonovich

The two integrals differ only in where the integrand is evaluated inside a step. Figure 1 shows the sample path and the two evaluation points; figure 2 shows what the choice accumulates to.

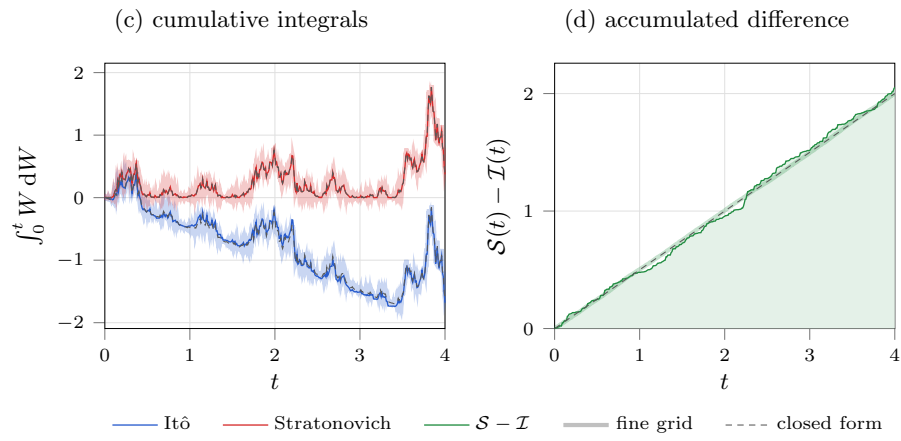


Figure 2: (c) The two readings of $\int_0^t W \, dW$, each shown on the integration grid and, pale behind it, on the fine grid, against the closed forms $(W^2 - t)/2$ and $W^2/2$. The Stratonovich sum reproduces the chain rule on any grid; the Itô sum sits off its closed form by $\frac{1}{2}(\sum \Delta W^2 - t)$. (d) The gap the two accumulate, against $t/2$.